Safari Research Proposal

Version 2 | 19 September 2026 | SD5008 Research and Analysis for Design | ISD

Group Name

Safari

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Research Project Title

Attention Allocation and Navigation Information Design: for Newcomers to Hong Kong

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Research Background

This study investigates how newcomers to Hong Kong obtain, interpret and act on navigation information while walking through unfamiliar surroundings. We aim to classify information in relation to the task it supports and the interaction demands it creates, then derive guidance on when and how it should be communicated. Walking navigation includes orientation, movement, stopping to check, finding an entrance and recovering from uncertainty.

Attention and situated interaction

Multiple resource theory explains why concurrent activities can interfere when they draw on overlapping processing resources; changing sensory modality does not remove all competition (Wickens, 2008). Oulasvirta et al. (2005) found fragmented attention during mobile web-search tasks across urban situations. Their findings motivate studying short interaction episodes, but do not establish a universal four-second attention limit or directly describe navigation. Wobbrock (2019) identifies how situational conditions can constrain technology use. Together, these perspectives suggest examining interface demands alongside environmental and bodily conditions, rather than treating information quantity alone as the source of difficulty.

Navigation information and burden

Cheng et al. (2023) compared maps showing three, five and seven landmarks during virtual navigation and found greater blink-related neural indicators of cognitive load in the seven-landmark condition. The study links information display with cognitive demands, but does not establish a universal landmark limit. Bartling et al. (2026) studied 54 participants in virtual environments: heavier traffic increased reported cognitive workload, while hiding task-irrelevant points of interest provided only partial support for navigation. These findings make the match between information, task and context a stronger starting point than a simple rule to display less.

In a real-world study of 33 adults, Shimokihara et al. (2025) found that smartphone glances were positively associated with stops and route errors. The authors could not establish causal direction: repeated glances might contribute to difficulty or represent attempts to resolve it. This distinction is central to our study. We will record event sequences and ask what participants were trying to confirm, rather than interpreting every phone check as excessive burden.

Alternative delivery and the proposed contribution

PocketNavigator associated tactile feedback use with less touchscreen interaction and fewer display checks in everyday use, but its observational deployment did not isolate causal effects on workload (Pielot et al., 2012). Nagata et al. (2026) compared voice navigation with paper maps in adults aged 50–69; relaxation improved with voice guidance, while fatigue and navigation performance did not significantly improve. The population and comparator limit direct transfer to our project. These studies establish relevant precedents without justifying a predetermined voice or haptic solution.

The proposed contribution is an episode-based account of how information roles, task demands and presentation conditions combine in newcomers' walking navigation. We will develop a contextual classification and a preliminary design decision framework from that account. The reviewed studies support this inquiry but do not prove that no similar classification exists. Novelty will be framed as a situated extension of existing work, with scope and transferability stated explicitly.

Research Objectives

- Identify information needs and observable interaction episodes across an unfamiliar walking journey.
- Explain when information access, interpretation, confirmation and action are experienced as effortful, and when additional information provides useful support.
- Develop a classification linking information roles, contextual demands and participant explanations, including contrasting cases.
- Derive evidence-linked guidance for presentation timing, detail and delivery, with clear limits on what has been evaluated.

Research Questions

Main question: How can navigation information be classified by its associated interaction burdens to inform interaction design methods for improving when and how it is presented and delivered?

The main question is retained from the group's registered topic. We interpret burden as contextual, rather than a fixed property of an information category.

- RQ1: What information do newcomers need at different stages of walking navigation, and how do they coordinate phone use with environmental attention?
- RQ2: Under what conditions do information presentation and task demands lead to effort, uncertainty, interruption or effective support?
- RQ3: How can these findings inform a classification and design guidance concerning information timing, detail and delivery?

Data Collection Methods

Research approach and scope

We propose a focused qualitative field study informed by attention theory. Observation will document actions and context; semi-structured interviews will investigate participants' interpretations. Descriptive event counts will support this analysis without turning it into a hypothesis-testing mixed-methods study. The design adopts the observational and interview approaches discussed in W03. It does not claim to be a full ethnography, a physiological study of attention, or a controlled test of cognitive mechanisms.

The analysis unit is an information-interaction episode: a sequence linking a need or trigger, information sought or received, user action and outcome. Interaction burden provisionally means the effort or disruption involved in obtaining, interpreting, retaining or acting on information. It will be inferred cautiously from contextual evidence and participant accounts, not diagnosed from pauses or glances alone. Cognitive load, uncertainty and fatigue will remain distinct concepts; we will not directly measure physiological load or fatigue.

Question	Evidence needed	Collection and analysis
RQ1	Information needed, visible actions and journey stages	A short profile interview, observed route and episode timeline.
RQ2	Triggers, competing demands, intended purpose and consequences	Low-intervention observation followed by episode-led semi-structured interviewing; compare explanations with actions.
RQ3	Relationships between information roles, context and preferred support	Brief card-assisted elicitation within the debrief, then a cross-case matrix linking evidence to design propositions.

Preparation and pilot

Before participant research, we will prepare the protocol, information sheet, consent form and route plan for the required ethics review. Researchers will scout one public route, normally taking about 15–20 minutes, from a transport exit to a public destination entrance. A backup route will be used only when familiarity, access or safety makes the primary route unsuitable; route differences will be documented rather than treated as equivalent experimental conditions. One eligible pilot volunteer, excluded from the main sample, will test the procedure after clearance.

One integrated session per participant

Each main session will take approximately 45–60 minutes: 5–10 minutes for consent and a short profile interview; 15–20 minutes for the observed walk; and 25–30 minutes for the debrief, including the card activity. There will be no required second visit.

Participants will use their usual navigation application and settings, at their own pace, without a speed target or continuous think-aloud task. One researcher will accompany them and a second will take timestamped notes. Researchers will avoid unsolicited conversation during navigation. We will record visible phone checks, stops, reorientation, requests for help and route corrections, together with the information visible or subsequently confirmed and the surrounding conditions. Uncertain observations will be marked as such. This is consented, low-intervention observation on an assigned task, not covert observation or precise gaze measurement.

The debrief will start with an open account of the journey, then revisit approximately three to five episodes, including both difficulty and smooth progress. Neutral prompts will include “What were you trying to find out here?”, “What happened just before you checked?” and “What helped you decide what to do next?”. We will ask what made an episode easy or demanding and whether extra information helped. The timeline will support immediate recall, while recognising that accounts remain retrospective interpretations.

In the final five minutes of the debrief, participants will write or confirm four to six information items from their own journey on blank cards. They will indicate where or when each item was needed, add missing items, and discuss what could wait or remain available on request. These prompts adapt the tangible elicitation tools introduced in W03, page 27. The activity supports expression and comparison; it is not a standalone co-design workshop, a validated taxonomy test or a vote determining the final classification.

Only the seated or stationary debrief will be audio-recorded, with separate consent. Participants may instead choose written notes. We will not require continuous screen recording, video or GPS logging. Researchers will not deliberately induce distraction, confusion or hazardous conditions; the walk can be paused or ended at any time.

Analysis and synthesis

We will adapt the Framework Method, which organises qualitative material into a structured matrix while retaining links to cases and source material (Gale et al., 2013). Each recording will be transcribed and checked; relevant statements will be linked to observation timestamps. We will familiarise ourselves with each complete case, code the first two cases openly, and refine a shared analytical framework before charting the remaining cases. Two researchers will discuss their independent interpretations of those first cases and retain an audit trail of revisions. Tutor feedback will support the team’s methodological learning.

For this project, the matrix will record participant, journey stage, trigger, information role, presentation conditions, visible action, participant explanation, outcome and evidence confidence. Orientation, action guidance, confirmation and recovery are sensitising categories rather than predetermined findings. Analysis will examine both within-journey sequences and differences between participants. An episode with weak evidence will remain unresolved; multiple episodes from one person will not be treated as independent participants.

Design propositions will follow an explicit chain: observed episode → participant explanation → contextual interpretation → proposed information support. Each proposition will include a supporting example, a counterexample or boundary condition, and its evaluation status. We will review two anonymised cases against the emerging framework internally. This checks traceability, not effectiveness with new users. A prototype trial or repeat participant session is outside the core plan and would require a separate rationale, time allocation and any necessary ethics amendment.

Who the Participants Are

We will plan six core participants, with up to two additional participants only if evidence gaps and the time budget justify them, plus one separate pilot. Participants will be adults who moved to Hong Kong within the previous six months, use smartphone navigation, are unfamiliar with the selected route, and can comfortably undertake the walk. English or Mandarin will be supported. Time since arrival is an operational recruitment criterion, not a validated boundary of navigation competence. Route familiarity, tool experience and language familiarity will be recorded separately.

Recruitment will seek some variation in navigation habits without creating subgroup quotas or a newcomer-versus-local comparison. Information power provides a basis for reviewing sample adequacy in relation to the study aim, specificity, dialogue quality, theory and analysis (Malterud et al., 2016); it does not prescribe six or eight participants. After six sessions, we will assess missing perspectives and thin evidence before deciding whether to invite up to two additional participants within the approved scope and time budget. If fewer cases or limited evidence are obtained, we will narrow the claims and report the limitation instead of asserting saturation.

How Participants Will Be Recruited

We will use purposive convenience recruitment through university and newcomer networks where invitations are permitted, with voluntary peer referrals. Interested people will contact the team privately. A short screener will check eligibility and route familiarity. We will avoid recruitment pressure, keep contact details separate and make clear that participation has no effect on grades or services. No payment is currently budgeted. A likely student-heavy sample will be acknowledged rather than described as representative of all newcomers.

Participant Information and Consent

Participants will receive an English or Chinese information sheet before scheduling and again at the session. It will explain the study purpose, activities, time, walking conditions, recording choices, data handling, withdrawal and researcher/supervisor contacts. Written or electronic consent will precede data collection; permission for audio and de-identified quotations will be addressed separately. The pilot is also subject to prior ethics clearance, as required in W03, page 53.

Participants may decline questions or recordings, pause or stop without penalty. We propose a seven-day window after each session for requesting removal of identifiable data; irreversible anonymisation will occur only after that window, subject to the approved protocol. Recruitment materials will contain a confirmed student contact and supervisor details. Unsafe weather, crowding or discomfort will lead to a pause, relocation or rescheduling; no research conversation will be required while crossing roads.

Confidentiality and Data Management

Participants will receive codes such as P01. Consent and contact records will be separated from transcripts and episode matrices in institution-approved, access-restricted storage. Only the research team and authorised supervisors will have access. Reports will remove names, home addresses, account information and potentially identifying combinations of details. Private messages, payment information, bystander identities and location histories will not be collected.

Audio will be used for checked transcription and deleted according to the approved retention plan. The named storage location, retention period and contact address will be confirmed with Dr. Rui He before recruitment and stated in the information sheet. Raw participant data will not be submitted to public AI services. Any transcription technology must be approved and disclosed in consent materials. The data will be confidential and de-identified in reporting, but collection will not be described as fully anonymous.

Anticipated Data Types

- Eligibility and contextual profiles, including time in Hong Kong, route familiarity and navigation habits.
- Timestamped field notes and whole-journey event timelines, including smooth episodes and interruptions.
- Consented audio or interview notes, checked transcripts and participant-created information cards.
- Descriptive counts and approximate timings, coded case matrices and analytic/reflexive memos.
- An evidence-linked information classification, design propositions and illustrated boundary cases.

Research Timeline

The plan follows the 2026 course brief and its five presentations. Dates below are planning targets except the published course milestones. Participant activities remain conditional on prior ethics clearance. If clearance is delayed, we will present the prepared workflow and document the delay instead of collecting unapproved data.

Date or period	Course milestone	Planned work and evidence
19–24 September	Presentation 1 on 24 September	Finalise topic, literature map, questions, method rationale and ethics submission materials; no empirical findings claimed.
25 September–7 October	1 October has no class; data gathering is scheduled to begin	Seek clearance, scout route and rehearse as researchers. After clearance, complete one pilot and, if feasible, two main sessions.
8 October	Presentation 2	Show research workflow, protocol revisions and available pilot/main-study documentation; state actual approval and recruitment status.
9–28 October	Tutorial 15 October; analysis lecture 22 October	Complete the remaining main sessions toward six to eight in total, normally one or two per week. Transcribe, summarise and code alongside collection.
29 October	Presentation 3	Show updated workflow, data inventory, inclusion/exclusion decisions and examples of filtering and coding. Aim to close core recruitment.
30 October–12 November	Synthesis lecture 5 November; Presentation 4 on 12 November	Build the cross-case matrix; develop and challenge categories; present emerging synthesis, uncertainty and counterexamples.
13–26 November	Findings tutorial 19 November; Presentation 5 on 26 November	Refine evidence-linked design propositions and limitations; demonstrate the framework with cases; prepare a substantial report draft.
27 November–6 December	Individual written assessment 3 December; final report due 6 December	Protect individual assessment preparation time; revise and verify the 5,000–8,000-word English report, APA 7 references and required declarations; submit PDF under 10 MB with plagiarism report appended.

Workload and division of work

The core plan comprises one pilot, six to eight integrated sessions and one analysis cycle. Four students will work in two pairs, each responsible for three core participants and, if justified, one additional participant. Pair members will rotate observation and interview roles and cross-check an example from the other pair. Recruitment, transcription and analysis will proceed in small batches to avoid a backlog.

The agreed planning basis is three hours per student per week, or twelve team person-hours. Across eleven weeks, the budget is 132 person-hours: 16 for literature/source checking; 14 for protocol, recruitment, route and pilot; 22 for field sessions, local travel and immediate notes; 22 for transcription and coding; 12 for synthesis; 14 for presentations; 20 for the report; and 12 contingency hours. This equals 33 hours per student and excludes class attendance and individual assessment preparation. Actual time will be logged weekly; optional work will be dropped or reassigned within the weekly budget if core tasks take longer.

If capacity is lower, we will prioritise six well-documented sessions, one route and a narrower set of information decisions. The card activity can be omitted if it adds little after the pilot. Large surveys, focus groups, longitudinal diaries, equipment-based attention measurement and prototype comparisons are not required deliverables. Extra methods will not be added merely because the calendar contains spare weeks.

Expected Outcome

We expect to produce an account of information needs and interaction demands across unfamiliar walking navigation, a contextual classification, and a preliminary design framework. The framework will connect task and information role with contextual demands, timing, level of detail, delivery considerations and access to confirmation or recovery. Illustrated cases will show how each recommendation follows from evidence, including cases where additional information helps.

The contribution is a structured way to reason about navigation information design. Its completeness, usability by other designers and effectiveness in reducing burden will remain unvalidated without further work. A small convenience sample, assigned route, observer presence, retrospective accounts and variation between apps constrain interpretation. The study cannot isolate effects of newcomer status, infer neural attention processes or generalise to driving, cycling or all Hong Kong newcomers.

Key References

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Appendix on Course Sources and Generative AI Use

Course sources: Course brief_SD5008, dated 3 September 2026, especially pages 2–6; W01_lecture; W02_lecture, especially the scope, research-question and research-through-design discussions; W03_lecture, especially pages 9–11, 19, 24–31 and 53; and the research proposal checklist. SAMPLE_research proposal_01, _02 and _03 informed the organisation of questions, methods and timeline, not the scientific claims or current deadlines.

Generative AI used: OpenAI ChatGPT/Codex. It assisted with organising the proposal, reading supplied course materials, searching and checking references, comparing method options, estimating workload, drafting English prose and formatting the document. No participant data have been collected or analysed in this proposal. The course brief requires disclosure of the tool and prompts used. The team must critically review the proposed methods, source interpretations and workload before submission; that review is not represented as already completed.

Prompt for the first draft, as provided by the user:

research proposal template topic research proposal

Prompt for this revision, as provided by the user:

research proposal sample research proposal reference

lecture slides

The wider conversation also included topic development, scope discussion and translation. Retain that conversation as the fuller prompt record, and add any subsequent AI-assisted revisions and their prompts to the final declaration.